Hypothesis testing with e-values

Chapter 9: FDR control using compound e-values (part 1)

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Recap & roadmap

Relevant concepts:

- P-variable $P \in [0, \infty]$: $\mathbb{P}(P \leq \alpha) \leq \alpha$ for all $\alpha \in [0, 1]$ and $\mathbb{P} \in \mathcal{P}$ (**Chapter 1**)
- E-variable $E \in [0, \infty]$: $\mathbb{E}_{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$ (**Chapter 1**)
- *p*-to-*e* and *e*-to-*p* calibrators (**Chapter 2**)

Outline:

- Multiple testing setup $\to$ FDR and limitations of p-value-based methods
- e-BH $\to$ post-hoc FDR control via compound e-variables
- Universality $\to$ connections to FDR procedures

Setup

Multiple hypothesis testing with K nulls $\{\mathcal{P}_i\}_{i\in[K]}$ and data $X \sim \mathbb{P}^\dagger$

- Multiple testing procedure $\mathcal{D} \colon \mathcal{X} \mapsto 2^{[K]}$ returns set of rejected hypotheses (discoveries) $\mathcal{D}(X) \subseteq [K]$

with **FDR control**

- Index set of true nulls: $\mathcal{N}(\mathbb{P}^\dagger) = \{k \in [K] \colon \mathbb{P}^\dagger \in \mathcal{P}_k\}$
- Total discoveries: $R_{\mathcal{D}} = |\mathcal{D}|$
- False discoveries: $F_{\mathcal{D}} = |\mathcal{N}(\mathbb{P}^\dagger) \cap \mathcal{D}|$
- False discovery proportion: $F_{\mathcal{D}}/(R_{\mathcal{D}} \vee 1)$
- False discovery rate (FDR) under $\mathbb{P}^\dagger$: $\mathrm{FDR}_{\mathcal{D},\mathbb{P}^\dagger} = \mathbb{E}_{\mathbb{P}^\dagger}[F_{\mathcal{D}}/(R_{\mathcal{D}} \vee 1)]$

Procedure $\mathcal{D}$ has FDR control $\alpha \in (0,1)$ under $\mathcal{P} := \bigcup_{k\in[K]} \mathcal{P}_k$ if

$$\mathbf{FDR}_{\mathcal{D},\mathbb{P}} \leq \boldsymbol{\alpha} \text{ for all } \mathbb{P} \in \mathcal{P}.$$

Remark: standard procedures often achieve the sharper control $(K_0/K)\alpha$ where $K_0 = |\mathcal{N}(\mathbb{P})|$; notice $\mathbb{P}^\dagger \notin \mathcal{P} \Rightarrow \mathrm{FDR}_{\mathcal{D},\mathbb{P}^\dagger} = 0$.

BH & BHY procedures

Benjamini-Hochberg (**BH**) procedure for $\alpha \in (0,1)$:

- Input: $p_1, \ldots, p_K$ p-values for $\{\mathcal{P}_i\}_{i \in [K]}$
- Sort from smallest to largest: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(K)}$
- Find

$$k^* = \max\{k \in [K] \colon p_{(k)} \leq \alpha k / K\}$$

- Reject all nulls with $p_i \leq p_{(k^*)}$

Benjamini-Yekutieli (**BHY**) procedure:

- Apply BH to $\{\ell_K p_i\}_{i \in [K]}$ where $\ell_K = \sum_{i=1}^{K} 1/i$.

Results:

- BH controls FDR at level $(K_0/K)\alpha$ under independence or PRDS
- BHY controls FDR at level $(K_0/K)\alpha$, but more conservative

Glossary: PRDS = positive regression dependence on a subset; think positively correlated p-variables.

Compound e-variables

Naive extension:

- Given e-values $e_1, \ldots, e_K$, apply BH to $p_i = 1/e_i$ (calibration)
- This works by e-to-p calibration, but we can be more general by using *e*-variables that are "*valid on average*"

$\mathbf{E} = E_1, \ldots, E_K \in [0, \infty]$ are compound e-variables for $\{\mathcal{P}_i\}_{i \in [K]}$ (under $\mathcal{P}$) if

$$\sup_{\mathbb{P} \in \mathcal{P}} \sum_{k \in \mathcal{N}(\mathbb{P})} \mathbb{E}_{\mathbb{P}}[E_k] \leq K$$

and they are called tight in case of equality.

Standard examples are (weighted) e-variables, but definition allows for non-trivial dependence structures:

- Convex combinations of compound e-variables $\{\mathbf{E}^{(\ell)}\}_{\ell=1}^{L}$
- Combining evidence from selective follow-up studies

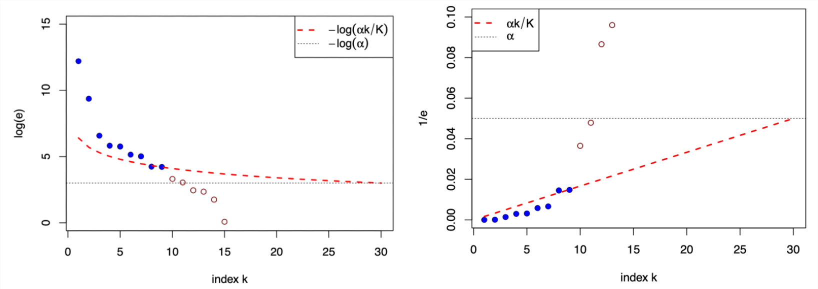
Outcome: adapting BH idea to compound e-variables enables FDR control under arbitrary dependence without additional corrections.

e-BH procedure for $\alpha \in (0,1)$:

- Input: $e_1, \ldots, e_K$ compound e-values for $\{\mathcal{P}_i\}_{i \in [K]}$
- Sort from largest to smallest: $e_{[1]} \geq e_{[2]} \geq \cdots \geq e_{[K]}$
- Find

$$k^* = \max\left\{k \in [K] \colon e_{[k]} \geq \frac{K}{\alpha k}\right\}$$

- Reject all nulls with $e_i \geq e_{[k^*]}$



Post-hoc FDR control of e-BH

For $\alpha \in (0,1)$, let $\mathcal{D}_\alpha$ be the e-BH procedure at level α. Then,

$$\mathbb{E}_{\mathbb{P}}\left[\sup_{\alpha\in(0,1)} \frac{F_{D_\alpha}}{\alpha R_{D_\alpha}}\right] \leq 1 \text{ for all } \mathbb{P} \in \mathcal{P}.$$

Proof:

- e-BH is a self-consistent procedure, i.e., $\mathcal{P}_k$ rejected implies $e_k \geq K/(\alpha R_{\mathcal{D}_\alpha})$
- Fix $\mathbb{P} \in \mathcal{P}$ and let $\mathcal{N} = \mathcal{N}(\mathbb{P})$. Then, by self-consistency,

$$\frac{F_{D_\alpha}}{\alpha(R_{D_\alpha} \vee 1)} = \sum_{k\in\mathcal{N}} \frac{\mathbf{1}(k \in D_\alpha)}{\alpha(R_{D_\alpha} \vee 1)} \leq \sum_{k\in\mathcal{N}} \frac{\mathbf{1}(k \in D_\alpha)E_k}{K} \leq \sum_{k\in\mathcal{N}} \frac{E_k}{K}$$

- Taking expectation and supremum over α on both sides concludes the proof.

Remarks: if $E_1, \ldots, E_K$ are e-variables, control level is $\alpha(K_0/K)$; e-BH dominates all other self-consistent procedures.

Universality of e-BH (1)

Prescriptive connection to BH and BHY procedures:

- BH: apply e-BH to boosted e-variables (i.e., e-variables taking advantage of PRDS structure)
- BHY: apply e-BH to calibrated e-variables

Descriptive connection to all FDR procedures: if $\mathcal{D}$ controls FDR at level α, then

1. There exist compound e-variables $\mathbf{E}$ such that e-BH gives the same discoveries as $\mathcal{D}$.
2. If $\mathcal{D}$ is admissible ($\mathcal{D}_\alpha \subseteq \mathcal{D} \Rightarrow \mathcal{D}_\alpha \equiv \mathcal{D}$), then $\mathbf{E}$ can be chosen to be tight.

Universality of e-BH (2)

(1) There exist compound e-variables $\mathbf{E}$ such that e-BH gives the same discoveries as $\mathcal{D}$.

Proof:

- For $V_k = \mathbf{1}(k \in \mathcal{D})$, define

$$E_k = \frac{K}{\alpha} \frac{V_k}{R_{\mathcal{D}} \vee 1}$$

- $\mathbf{E} = (E_k)$ are compound e-variables since, for every $\mathbb{P} \in \mathcal{P}$,

$$\sum_{k \in \mathcal{N}} \mathbb{E}_{\mathbb{P}} \left[\frac{K}{\alpha} \frac{V_k}{R_{\mathcal{D}} \vee 1} \right] = \frac{K}{\alpha} \mathbb{E}_{\mathbb{P}} \left[\frac{F_{\mathcal{D}}}{R_{\mathcal{D}} \vee 1} \right] \leq K$$

- By self-consistency, $\mathbf{E}$ induces the same discoveries as $\mathcal{D}$ since

$$E_k = \begin{cases} K/(\alpha(R_{\mathcal{D}} \vee 1)) & \text{if } V_k = 1 \\ 0 & \text{if } V_k = 0 \end{cases}$$

Universality of e-BH (3)

(2) If $\mathcal{D}$ is admissible ($\mathcal{D}_\alpha \subseteq \mathcal{D} \Rightarrow \mathcal{D}_\alpha \equiv \mathcal{D}$), then $\mathbf{E}$ can be chosen to be tight.

Proof:

- Take $E_k = KV_k/(\alpha(R_{\mathcal{D}} \vee 1))$ again and let

$$K^* := \sup_{\mathbb{P}\in\mathcal{P}} \sum_{k\in\mathcal{N}(\mathbb{P})} \mathbb{E}_{\mathbb{P}}[E_k]$$

- If $K^* = K$, done!
- If $K^* < K$:
    - If $K^* = 0$, then $\mathcal{D} \equiv \emptyset$ and can take $\mathbf{E} = \mathbf{1}$
    - If $K^* > 0$, define

$$E'_k = \frac{K}{K^*} E_k \text{ for } k \in [K]$$

        - $\mathbf{E}' = (E'_k)$ are tight compound e-variables and control FDR at level α
        - Since $K/K^* > 1$, $\mathbf{E}'$ induce at least as many discoveries as $\mathcal{D}$
        - By admissibility of $\mathcal{D}$, e-BH with $\mathbf{E}'$ give same discoveries as $\mathcal{D}$

Combining discoveries across procedures

While mainly a theoretical result, e-BH universality suggests a way to combine discoveries from different FDR procedures:

- Suppose we have run $\{\mathcal{D}^{(\ell)}\}_{\ell=1}^{L}$ at levels $\{\alpha_\ell\}_{\ell=1}^{L}$
- For each $\mathcal{D}^{(\ell)}$, construct compound e-variables $\mathbf{E}^{(\ell)}$ that induce the same discoveries
- Combine the e-variables via convex combination:

$$E_k = \sum_{\ell=1}^{L} w_\ell E_k^{(\ell)} \text{ with } w_\ell \geq 0, \sum_{\ell=1}^{L} w_\ell = 1$$

- Apply e-BH at level α to $\mathbf{E} = (E_k)$ to get combined discoveries

Remarks: the value of α_ℓ are implicitly used in the construction of $\mathbf{E}^{(\ell)}$; can be useful to derandomise procedures.

Experiment

- **Goal**: detect cryptocurrencies with positive expected returns
- **Challenge**: asset returns are heavily dependent
- **Setup**: $K = 100$ to 496 assets, daily returns $\{r_{k,t}\}_{t=1}^{T}$ for $T = 78$
  - Null $\mathcal{P}_k$: $\mathbb{E}[r_{k,t}|\mathcal{F}_{t-1}] \le 0$ for $t = 1, \ldots, T$
  - E-variable: $E_k = W_{k,T}$, where $W_{k,t} = \prod_{s=1}^{t}(1 + \lambda r_{k,s})$ with $W_{k,0} = 1$
  - P-variable: $P_k = (\max_t W_{k,t})^{-1}$
  - Strategies: $\lambda \in \{1, 0.5, 0.1\}$

Table 3: Number of coins selected by e-BH and BY procedures

	buy and hold			50%-rebalancing			30%-rebalancing		
#total coins	496	200	100	496	200	100	496	200	100
e-BH 5%	3	4	0	42	28	13	21	12	6
BY 5%	11	0	0	15	11	6	15	4	5
e-BH 10%	13	7	2	60	48	24	32	22	8
BY 10%	18	7	0	25	15	7	13	7	7

Takeaways & what comes next

Main takeaways (first half of Chapter 9):

- **Compound e-values**: relaxation of e-values that can accommodate complex dependence
- **e-BH procedure**: natural extension of BH to compound e-values giving post-hoc FDR control under arbitrary dependence
- **Universality**: every procedure with FDR control can be represented as e-BH on special compound e-values

Preview (second half of Chapter 9):

- **Optimality**: construction of powerful compound e-values
- **Improvements over basic e-BH**: randomisation and closure